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Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Evaluate the following integral using integration by parts.

$$\int_1^3 16x \ln x \, dx$$

The general form for integration by parts is $\int u \, dv = uv - \int v \, du$.

Identify the potential expressions for u and dv . Then calculate du and v .

If $u = \ln x$ and $dv = 16x \, dx$, then $du = \frac{1}{x} dx$ and $v = 8x^2$.

Apply integration by parts.

$$\begin{aligned} \int_1^3 16x \ln x \, dx &= \left[8x^2 \ln x \right]_1^3 - \int_1^3 8x \, dx \\ &= \left[8x^2 \ln x \right]_1^3 - \left[4x^2 \right]_1^3 \\ &= 72 \ln 3 - 32 \end{aligned}$$